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REVIEW ARTICLE



A systematic review of ionizing radiation-induced glaucoma: clinical manifestations, pathogenesis, and current treatment approaches

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ABSTRACT

Purpose: Ionizing radiation-induced glaucoma (IRG) is an uncommon yet vision-threatening complication that presents primarily as neovascular glaucoma (NVG) after high-dose exposure and normal-tension glaucoma (NTG) after low-dose exposure. Despite increasing recognition, its dose-response relationship, mechanisms, and optimal management remain poorly defined. This review evaluates the clinical spectrum, pathogenesis, and therapeutic approaches of IRG to inform future research and practice.

Methods: A systematic literature search of PubMed/MEDLINE, Embase, Scopus, and the Cochrane Database (1964–2024) was conducted following PRISMA guidelines. Among 7256 screened articles, 37 studies fulfilled the inclusion criteria.

Results: NVG typically develops following ocular radiotherapy exceeding 30Gy, particularly in patients with uveal melanoma, and is characterized by retinal ischemia, neovascularization, and secondary intraocular pressure elevation. NTG arises from chronic low-dose exposure (< 1Gy), producing optic nerve injury and visual field loss without elevated pressure. Anti-VEGF therapy can temporarily control NVG progression, whereas trabeculectomy, glaucoma drainage devices, and cyclodestructive procedures are variably employed but often limited by postoperative complications and inconsistent outcomes. The lack of consensus-driven protocols and heterogeneity across studies underscore ongoing clinical challenges.

Conclusions: IRG represents a dose-dependent entity with distinct phenotypes and mechanisms. Current therapies provide partial benefit but remain unsatisfactory in terms of durability and standardization. Advancing the field will require mechanistic studies to clarify radiation-induced optic neuropathy and vascular injury, alongside well-designed trials to establish preventive strategies and evidence-based treatment algorithms.

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Introduction

Ionizing radiation plays a transformative role in modern healthcare and technology. From diagnostic imaging to cancer therapies, radiation has proven to be an invaluable tool [1,2]. However, as the use of radiation continues to increase, so do the associated health risks, particularly for individuals exposed to high levels of radiation in occupational settings. Among the most concerning adverse effects is radiation-induced damage to the ocular structures, including the lens and retina, which are particularly sensitive to ionizing radiation [3]. Such damage can lead to a variety of ocular conditions, with glaucoma emerging as one of the

most significant and potentially devastating complications [4].

Glaucoma refers to a group of optic neuropathies characterized by progressive degeneration of the optic nerve and irreversible visual field loss, often in association with elevated intraocular pressure (IOP) [5]. While glaucoma can manifest as a primary disease, ionizing radiation-induced glaucoma (IRG) is typically classified as a secondary form of the disease [6]. It is most commonly observed in patients who have received radiation therapy to ocular or adjacent tissues. The pathophysiology of IRG involves complex mechanisms, including damage to the retinal vasculature and the optic nerve, which contribute to elevated

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IOP and subsequent glaucomatous changes [7]. Neovascular glaucoma (NVG), the most frequent form of IRG, results from abnormal fibrovascular tissue proliferation in the iris and anterior chamber angle, leading to increased IOP and potential vision loss [8]. Normal-Tension Glaucoma (NTG) is mainly caused by radiation-induced mitochondrial dysfunction in retinal ganglion cells (RGCs) [9]. The senescence-associated secretory phenotype (SASP) in chronic radiation injury exacerbates the glaucomatous neuropathy beyond the vascular endothelial growth factor (VEGF)-mediated pathway [10].

The clinical management of glaucoma generally aims to reduce IOP in order to prevent further optic nerve damage and preserve visual function [11]. For IRG, treatment options can be challenging due to the unique pathogenesis of the condition. Common approaches include topical antiglaucoma medications, retinal laser therapies, and intravitreal injections, such as bevacizumab (an anti-VEGF agent), which can help control IOP and address neovascularization [8,12]. However, the effectiveness of these treatments is often variable, and no standardized treatment protocols currently exist, especially for more advanced stages of the disease.

While IRG remains an important clinical issue, the existing literature lacks a comprehensive review that consolidates the available evidence on its epidemiology, pathogenesis, and treatment strategies. This review aims to address this gap by providing an in-depth examination of IRG, with a particular focus on its clinical manifestations, pathophysiology, and the current therapeutic approaches. Through this comprehensive analysis, we seek to contribute to the development of more personalized and effective management strategies, ultimately improving patient outcomes and minimizing the impact of this challenging condition on those affected.

Materials and methods

Search strategies

The current systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for reporting systematic reviews (Figure 1). A comprehensive literature search was performed across several databases, including PubMed/MEDLINE, Embase, Scopus, and the Cochrane Database of Systematic Reviews, covering publications from 1964 to 2024. The search terms used included: "Glaucoma" and "Radiation" (Supplemental Tables 1–5).

Inclusion and exclusion criteria

For paper selection, studies were included if they focused specifically on IRG, excluding those that addressed other radiation-related diseases. Additionally, reviews, case reports, commentaries, and letters were not considered for inclusion. Only studies published in the English language were selected for review.

Data extraction and quality assessment

Two independent reviewers assessed the reliability of the clinical data and the subsequent treatment recommendations for patients with IRG. There is no single recommended instrument to assess the risk of bias for systematic reviews that includes both prospective and retrospective observational studies. We chose to evaluate the risk of bias by using a previously published system based on a 10-item assessment (Supplemental Figure 1). The system was developed to evaluate the kind of bias and the completeness of information. Particularly, items 1 and 2 assess the selection bias, items 3 to 5 the reporting bias, item 6 the attrition bias, and items 7 to 10 assess the extensiveness of information on therapeutic approaches and outcomes. Specifically, an affirmative answer, "yes," indicates a low risk of bias, while a negative answer, "no," a high risk. Unclear or unknown risks are indicated by the definition "unclear." Disagreements between the reviewers were resolved through mutual discussion and consensus.

Results

Types of ionizing radiation-induced glaucoma

Ionizing radiation has been shown to lead to various types of glaucoma, reflecting the multifactorial mechanisms through which radiation can affect ocular physiology. Two main types of IRG have been identified: neovascular glaucoma and normal-tension glaucoma (Figure 2) [13,14].

Neovascular glaucoma (NVG)

NVG is characterized by the growth of abnormal blood vessels in the iris or anterior chamber angle, leading to an increase in IOP and subsequent glaucomatous damage [15,16]. This form of glaucoma is often associated with ischemic conditions in the eye and has been reported following radiation therapy for uveal melanoma [17]. Ischemic conditions trigger the release of angiogenic factors, particularly VEGF, which promotes neovascularization, contributing to ocular closure [13].

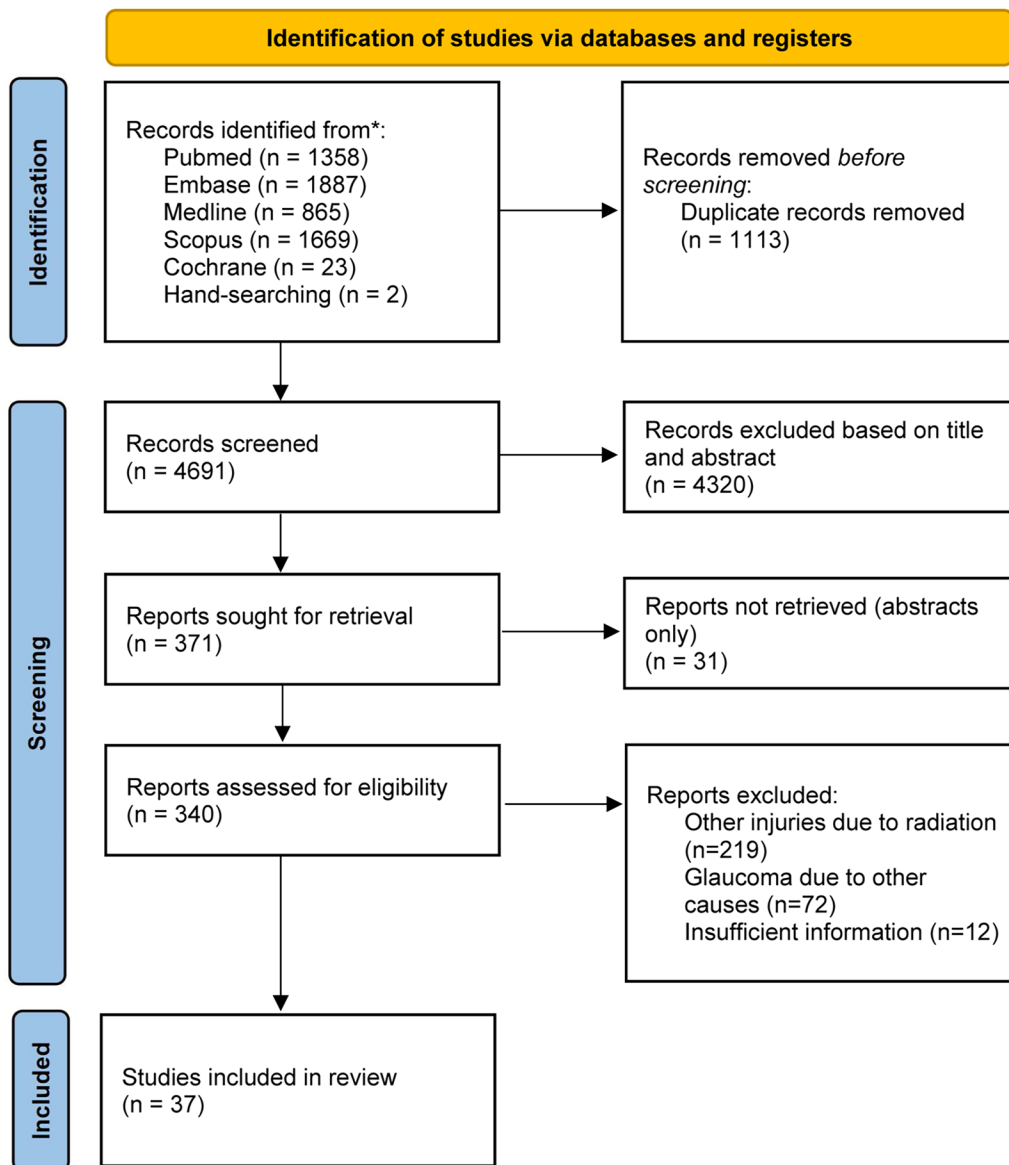


Figure 1. PRISMA flow diagram illustrating the study selection process. Initial records identified from multiple databases (PubMed, Embase, Medline, Scopus, Cochrane, and hand-searching) totaled 5,794. After removing duplicates ($n=1,113$), 4,691 records were screened based on titles and abstracts, excluding 4,320. Full-text retrieval was attempted for 371 reports, of which 31 were unavailable. Eligibility assessment of 340 reports excluded 303 due to unrelated injuries ($n=219$), glaucoma from other causes ($n=72$), and insufficient data ($n=12$). Finally, 37 studies were included in the review.

Radiation-induced NVG is believed to result from the upregulation of pro-angiogenic factors, primarily VEGF, due to radiation-induced tissue damage, leading to neovascularization and the subsequent elevation of IOP [18].

Normal-Tension Glaucoma (NTG)

NTG, on the other hand, is diagnosed in the presence of glaucomatous optic neuropathy despite IOP remaining within the statistically normal range [19]. In studies involving atomic bomb survivors, a significant

association was found between NTG and radiation exposure, suggesting that ionizing radiation may play a role in the development of this glaucoma subtype [20]. The pathogenesis of NTG is not yet fully understood but is thought to involve microvascular damage, ischemia, and possibly ischemia of the optic nerve head [21].

Incidence of ionizing radiation-induced glaucoma

Studies have demonstrated significant variability in the incidence of IRG, depending on the type and dose of

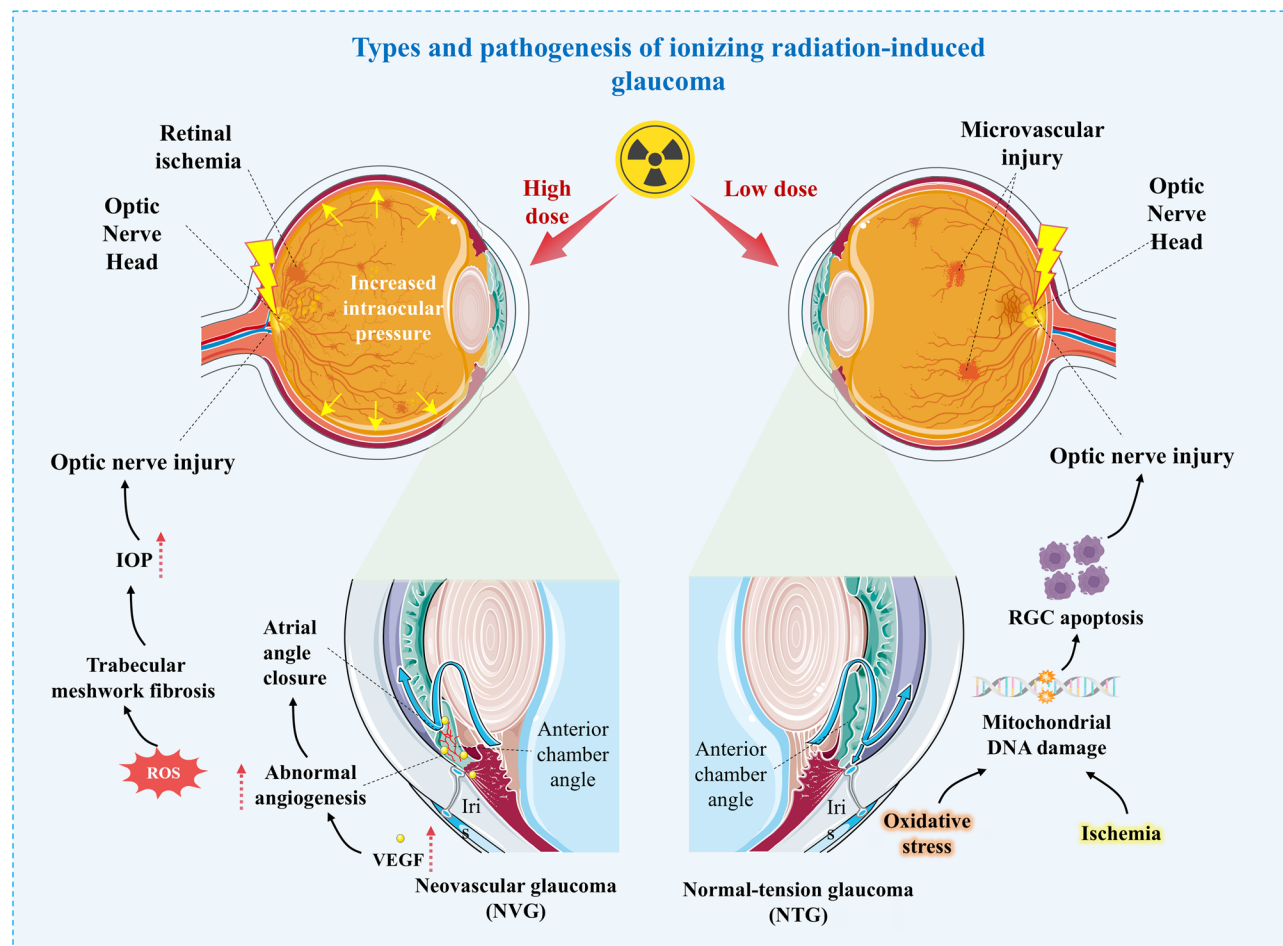


Figure 2. Types and pathogenesis of ionizing radiation-induced glaucoma. Schematic illustration of the types and pathogenesis of IRG. High-dose radiation causes retinal ischemia, increased intraocular pressure and optic nerve damage. Low-dose radiation causes microvascular injury and subsequent optic nerve damage. Pathological mechanisms include abnormal angiogenesis (VEGF-driven neovascular glaucoma) and oxidative stress-induced DNA damage (normal-tension glaucoma). RGCs apoptosis and ischemic effects also contribute to optic nerve degeneration. This figure highlights the different pathways underlying radiation-induced glaucoma subtypes.

radiation exposure (Table 1). A study on Mayak nuclear workers in Russia established a strong association between chronic ionizing radiation exposure and the development of NTG [22]. Workers who experienced gamma-ray exposure exceeding 1 Gy to the brain had a notably higher relative risk (RR) for NTG (RR = 1.88, 95% CI: 1.01, 3.51; $p=0.047$). Furthermore, a linear correlation was observed between the incidence of NTG and the brain's absorbed gamma dose, with an excess relative risk of 0.53 per unit dose (95% CI: 0.01, 1.68; $p<0.05$) [22].

A retrospective analysis examining the risk of NVG following proton beam therapy (PBT) for choroidal melanoma found that 20.8% of patients developed iris neovascularization, with 47% progressing to NVG approximately 2 years after treatment [13]. The incidence of neovascularization following PBT for choroidal melanoma ranged from 7.0% to 42.8%, depending

on factors such as tumor characteristics and radiation technique [24]. Further studies have reported varying incidences of secondary glaucoma in patients treated with radiation. For instance, Eun-Ah Kim *et al.* reported an 8.6% risk of secondary open-angle glaucoma and a 6.7% risk of NVG following iodine-125 brachytherapy for uveal melanoma [25]. A comparative study of robotic radiosurgery (RRS) and brachytherapy (BT) for uveal melanoma found that the incidence of secondary glaucoma was significantly higher at 5 years for patients treated with RRS (46.7%) compared to those treated with BT (11.1%) [26]. These findings highlight the complex and dose-dependent relationship between ionizing radiation and the development of glaucoma, emphasizing the need for careful monitoring of patients who undergo radiation therapy for ocular or adjacent tissues.

Table 1. Incidence of ionizing radiation-induced glaucoma.

Study Population	Type of Radiation Exposure	Radiation Modality	Glaucoma Type	Incidence Rate (%)	Relative Risk (RR)	Confidence Interval (CI)	Mean Time to Development (Years)	References
Choroidal Melanoma Patients (PBT)	Proton beam therapy	Proton beam therapy (PBT)	NVG	20.8	N/A	N/A	2.0	[18]
Uveal Melanoma Patients (Iodine-125 Brachytherapy)	Brachytherapy	Iodine-125 Brachytherapy (BT)	Secondary Open-Angle Glaucoma	8.6	N/A	N/A	N/A	[21]
Uveal Melanoma Patients (Iodine-125 Brachytherapy)	Brachytherapy	Iodine-125 Brachytherapy (BT)	NVG	6.7	N/A	N/A	N/A	[21]
Uveal Melanoma Patients (Robotic Radiosurgery vs. Brachytherapy)	Robotic Radiosurgery	Robotic Radiosurgery (RRS)	Secondary Glaucoma	46.7	N/A	N/A	>5	[22]
Uveal Melanoma Patients (Robotic Radiosurgery vs. Brachytherapy)	Brachytherapy	Iodine-125 Brachytherapy (BT)	Secondary Glaucoma	11.1	N/A	N/A	>5	[22]
Mayak Nuclear Workers	Chronic gamma ray exposure	Chronic occupational exposure (gamma)	NTG	N/A	1.88	1.01-3.51	N/A	[23]

NTG: Normal Tension Glaucoma; NVG: Neovascular Glaucoma; PBT: Proton Beam Therapy; RR: Relative Risk; CI: Confidence Interval; N/A: Not applicable or not available.

Table 2. Types and dosages of ionizing radiation associated with glaucoma development.

Type of Ionizing Radiation	Dosage (Gy)	Glaucoma Type	Incidence Rate (%)	Relative Risk (RR) or Odds Ratio (OR)	95% Confidence Interval (CI)	References
Chronic low-dose radiation (Mayak nuclear workers)	>1 Gy	NTG	N/A	RR: 1.88	1.01-3.51	[16]
High-dose radiotherapy	>30-40	NVG	7.0-42.8	N/A	N/A	[24]
Proton beam therapy (PBT)	N/A	NVG	12.7 (5-year)	N/A	N/A	[25]
Carbon ion radiotherapy	N/A	NVG	42.6 (3-year)	N/A	N/A	[26]
Acute low-dose radiation (Japanese atomic bomb survivors)	<1 Gy	NTG	N/A	OR: 1.31	1.11-1.53	[27]
Low-dose ionizing radiation (US radiologic technologists)	<1 Gy	Glaucoma	N/A	N/A	N/A	[28]

NVG: Neovascular Glaucoma; NTG: Normal Tension Glaucoma; Gy: Gray (unit of absorbed dose); N/A: Not applicable or not available; RR: Relative Risk; OR: Odds Ratio; CI: Confidence Interval.

Types and dosages of ionizing radiation that induce glaucoma

Ionizing radiation has been strongly implicated in the development of glaucoma, particularly in the context of high-dose fractionated radiation therapy (Table 2). High-dose radiotherapy, commonly used in the treatment of uveal melanoma, is a well-established risk factor for the development of NVG [1]. Reported incidence rates of NVG following radiotherapy range from 7.0% to 42.8%, with doses typically exceeding 30–40 Gy [27]. PBT, a form of external radiotherapy, has shown a 5-year cumulative NVG rate of 12.7% [23]. Similarly, carbon ion radiotherapy, another high-dose treatment, has a 3-year cumulative NVG rate of up to 42.6%, highlighting the dose-dependent nature of radiation-induced glaucoma [28].

In contrast, the effects of lower doses of ionizing radiation on glaucoma development are less well studied. Low-dose or low-dose-rate irradiation can induce adaptive cellular protection mechanisms, including upregulation of antioxidant enzymes, enhanced DNA repair, and inhibition of inflammatory pathways, thereby mitigating to some extent the damage caused by subsequent high-dose radiation or other stress factors [29–31]. A study of Japanese atomic bomb survivors exposed to lower acute doses found a significantly increased risk for NTG at doses much lower than those used in radiotherapy [28,32]. Kiuchi *et al.* reported an odds ratio of 1.31 per 1 Gy for NTG prevalence (95% CI: 1.11, 1.53, $p=0.001$) [20]. On the other hand, a study of U.S. radiologic technologists exposed to low-dose ionizing radiation found no significant association between radiation dose and self-reported glaucoma,

suggesting a negligible risk at these levels [33]. However, data from Russian Mayak nuclear workers, chronically exposed to lower doses, revealed a significant increase in NTG risk with gamma-ray exposure exceeding 1 Gy to the brain. These findings align with those observed in Japanese atomic bomb survivors, suggesting that long-term, low-dose radiation exposure may also contribute to the development of NTG [22].

In summary, while high-dose radiotherapy is a recognized cause of NVG, emerging evidence suggests that even low-dose ionizing radiation, particularly with prolonged exposure, can contribute to glaucoma development, particularly in the form of NTG.

Pathogenesis of ionizing radiation-induced glaucoma

Ionizing radiation exposure triggers complex, multifactorial processes that contribute to the development of glaucoma by affecting key ocular structures, particularly the retina and optic nerve head (Figure 2) [22]. Retinal damage from radiation can lead to neovascularization, which plays a central role in the pathogenesis of NVG [13]. This process is thought to result from an imbalance in angiogenic factors, notably the upregulation of VEGF. Increased VEGF levels in the aqueous humor of patients with uveal melanoma and NVG are commonly observed [34]. Radiation-induced retinopathy, characterized by vascular damage and ischemia, further stimulates VEGF release, promoting the growth of abnormal blood vessels [35]. These vessels obstruct the iridocorneal angle, thereby increasing IOP and contributing to the onset of glaucoma [21].

Direct damage to the optic nerve head is another critical factor in the development of NTG, a form of glaucoma characterized by progressive optic nerve damage despite normal IOP [25]. This pathogenesis is associated with ionizing radiation's effects on RGCs and other ocular cells [36]. Ionizing radiation triggers DNA damage, leading to double-strand breaks and genomic lesions, a process known as the DNA damage response (DDR) [37]. DDR pathways activate cell cycle checkpoints and apoptosis, which can accelerate RGCs loss and contribute to glaucomatous optic neuropathy [38]. The accumulation of DNA damage in RGCs, alongside chronic cellular stress, may lead to optic nerve head degeneration over time [39]. Chronic hypoxia and mitochondrial dysfunction in RGCs lead to downregulated ATM kinase expression, impairing DNA double-strand break repair. Inactivation of the ATM-Chk2-p53 pathway results in persistent DNA damage, γ H2AX accumulation, aberrant PARP activation,

and mitochondrial apoptosis. Concurrently, failure of cell cycle checkpoints prompts damaged cells to enter mitosis prematurely, exacerbating genomic instability. These processes ultimately cause progressive RGCs death and axonal degeneration, even in the absence of elevated intraocular pressure [40]. In addition to direct cellular damage, radiation-induced inflammation and oxidative stress are key factors in glaucoma pathogenesis. Ionizing radiation increases the production of reactive oxygen species (ROS), which can damage cellular components and exacerbate RGCs apoptosis, as well as impair trabecular meshwork function, leading to elevated IOP [41,42]. Radiation-induced oxidative stress also leads to damage of microvascular endothelium, and the decrease in blood flow causes ischemic injury to the optic nerve [43]. Moreover, radiation-induced inflammation triggers the activation of pro-inflammatory cytokines and chemokines, which can further contribute to the development of NVG by promoting abnormal vessel formation and damaging ocular structures [44,45].

Clinical management of ionizing radiation-induced glaucoma

The clinical management of IRG involves various therapeutic strategies aimed at mitigating elevated IOP and preventing further optic nerve damage (Figure 3). A commonly employed approach is the use of anti-VEGF therapy, which has demonstrated promising outcomes in preserving vision in patients with radiation optic neuropathy (RON) and radiation maculopathy (RM) [46]. In a prospective clinical case series, the anti-VEGF agent bevacizumab was shown to significantly reduce the clinical evidence of RON in 100% of patients, with 64% maintaining stable or improved visual acuity over a median follow-up period of 29 months [47]. The recurrent neovascularization caused by the upregulation of compensatory angiopoietin-2 usually requires combined treatment, such as VEGF+Ang-2 inhibitors [48]. Trabeculectomy, a traditional surgical procedure designed to regulate IOP, has also been utilized in patients with IRG. Compared with the unirradiated eyes, fibrosis after radiotherapy reduces the success rate of trabeculectomy by approximately 40%, their success often depends on patient-specific factors and radiation dose [49]. A retrospective evaluation demonstrated a significant reduction in median IOP from 35 mmHg prior to surgery to 16 mmHg at six months postoperatively. Notably, no patients required glaucoma medications at the six-month follow-up [50]. Additionally, Baerveldt shunts have proven effective in lowering IOP with minimal complications in eyes

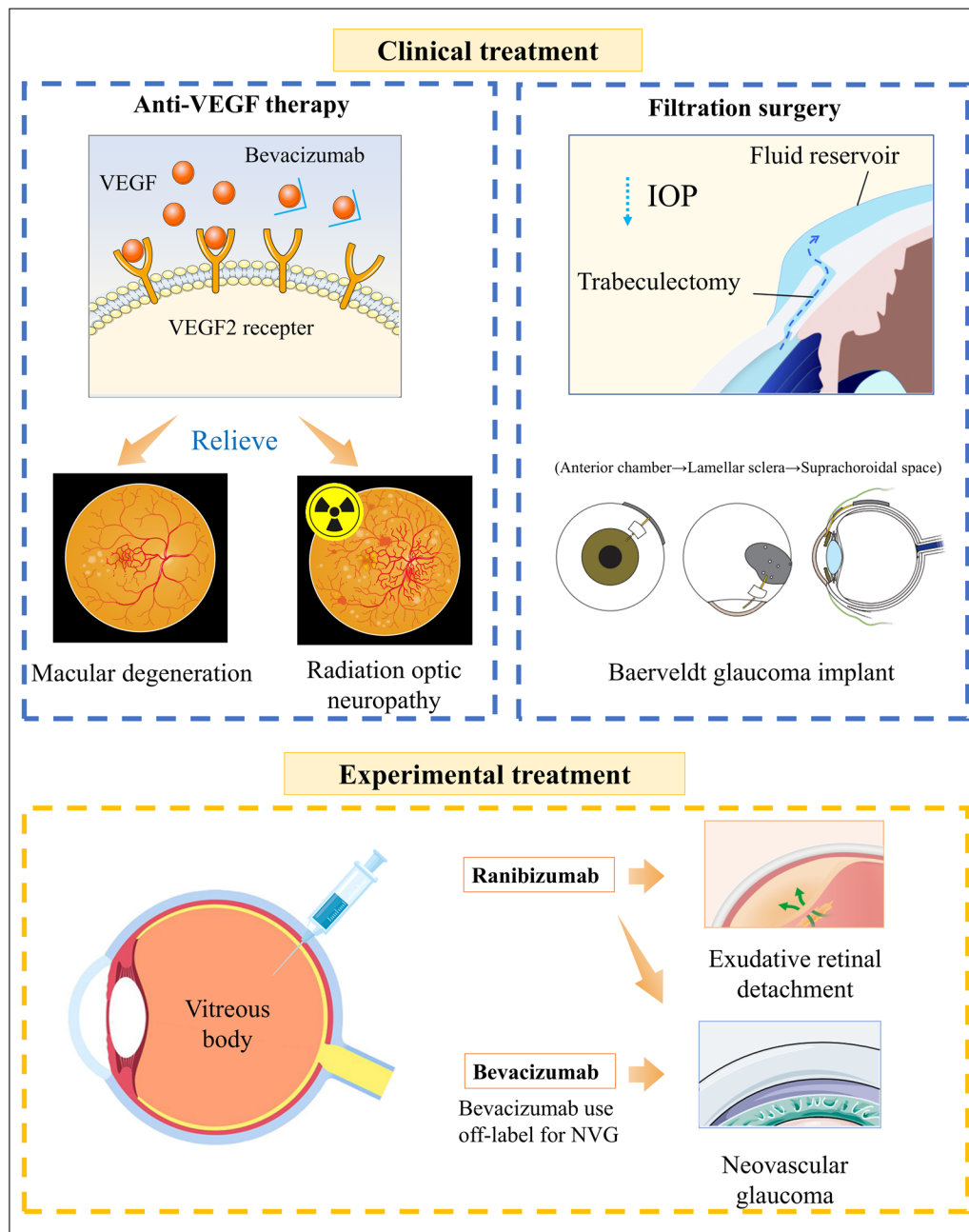


Figure 3. Clinical management of ionizing radiation-induced glaucoma. This diagram illustrates clinical and experimental treatments for eye diseases. Clinical treatments include anti-VEGF therapy and filtration surgery. Experimental treatment explores the use of ranibizumab and bevacizumab in the treatment of exudative retinal detachment and neovascular glaucoma. The figure highlights therapeutic mechanisms and potential applications in ophthalmology.

treated with total anterior segment irradiation for anterior uveal melanoma, suggesting that tube implantation could be a valuable option for managing radiation-induced glaucoma and preserving visual function [51].

Pharmacological interventions are also being explored in experimental settings to mitigate the effects of radiation-induced neovascularization and prevent further glaucoma progression. Intravitreal bevacizumab, an anti-VEGF agent, has been investigated

for its potential to reduce the enucleation rate and control IOP in patients with NVG following proton beam therapy for uveal melanoma. Studies suggest that intravitreal bevacizumab is effective in the early stages of NVG or as a preventive treatment [52]. Similarly, ranibizumab injections have been shown to rapidly improve NVG and exudative retinal detachment, further highlighting the potential of anti-VEGF therapies in treating radiation-induced neovascular complications [53].

Discussion

This systematic review underscores the intricate nature of IRG, emphasizing its multifactorial pathogenesis and varying prevalence rates. Two primary subtypes, NVG and NTG, dominate the clinical spectrum of IRG. High-dose radiotherapy, particularly in treating uveal melanoma, consistently correlates with a substantial risk of NVG, with incidence rates ranging from 7.0% to 42.8% [3,28]. Conversely, low-dose ionizing radiation, as evidenced in atomic bomb survivors and individuals with occupational radiation exposure, is strongly linked to an increased risk of NTG [20,32]. These findings reflect a dose-dependent relationship between radiation and glaucoma development, highlighting significant risks at both high and low exposure levels.

We also considered other confounding factors, such as East Asian ethnicity, female gender, Raynaud's phenomenon, and smoking history, which may be more causally significant than radiation in NTG [22,54]. China had the highest number of affected individuals among countries/regions in 2010. The prevalence rate of primary angle-closure glaucoma (PACG) was highest in the Chinese population, followed by the Japanese population, while European and Indian populations showed the lowest rates. East Asians exhibit a significantly higher probability of developing PACG compared to other regions [55]. Systemic vascular abnormalities such as nocturnal hypotension, low ocular perfusion pressure (OPP), migraines, and Raynaud's phenomenon are closely associated with the progression of NTG [56]. Heavy long-term smoking may significantly increase the risk of Primary Open-Angle Glaucoma (POAG) in certain populations [57].

Challenges in treatment and management

Managing IRG is notably complex due to its diverse pathophysiology, which encompasses vascular, inflammatory, and oxidative stress pathways. Anti-VEGF therapies, such as bevacizumab, have shown potential in mitigating neovascularization and preserving vision in patients with RON [46,47]. A prospective clinical case series demonstrated that bevacizumab significantly reduced signs of RON, with 64% of patients achieving stable or improved visual acuity over a median follow-up of 29 months [47]. However, the overall efficacy of anti-VEGF therapy in IRG remains contentious, as some studies have reported limited effectiveness in addressing optic nerve denervation [52].

Surgical interventions, including trabeculectomy and glaucoma drainage implants like Baerveldt shunts, have been employed to manage elevated IOP in IRG

patients [50,51]. While these approaches can effectively lower IOP, their success often depends on patient-specific factors and radiation dose. Moreover, these interventions primarily address the symptomatic aspect of glaucoma without targeting the underlying radiation-induced damage, leading to suboptimal long-term outcomes. The heterogeneity in treatment responses and the lack of standardized protocols for IRG further complicate clinical management. Variations in the type and dose of radiation, patient demographics, and baseline ocular health contribute to inconsistent therapeutic outcomes [58]. These inconsistencies underscore the need for more homogeneous study designs and standardized protocols to reliably evaluate treatment efficacy and improve patient outcomes.

Path forward: Research and prevention strategies

The gaps in current treatment approaches highlight the urgent need for targeted therapies tailored to the unique pathophysiology of IRG. Future research should focus on unraveling the molecular mechanisms underlying IRG, particularly the role of radiation in triggering neovascularization and optic nerve damage. Insights into these mechanisms can facilitate the development of novel therapeutic strategies, such as selective VEGF inhibitors or agents targeting oxidative stress pathways [59]. Preventive strategies are equally critical in mitigating the risk of IRG. For radiation practitioners and patients, strict adherence to radiation safety protocols, such as using leaded eyeglasses and appropriate shielding, is essential to minimize ocular exposure [60]. Regular ocular monitoring and early detection of glaucoma-related changes through comprehensive eye examinations can enable timely interventions, potentially preventing irreversible vision loss [61]. Technical improvements in radiation therapy also play an important role in preventing IRG. With the cross-development of radiation biology and physical technology, clinicians can now use a variety of methods to significantly reduce the radiation dose to the eye's accessory structures while maintaining therapeutic gains, avoiding indirect damage to the outflow structures, reducing retinal ischemia, and implementing personalized, anatomically specific radiation plans, thereby reducing the incidence of radiation-induced pathology.

Radiation protection agents, such as amifostine, have shown protective effects by counteracting ionizing radiation-induced hematopoietic and tissue damage [62]. According to the "Protection-Mitigation-Treatment" framework proposed by Brizel, radioprotectors aim to improve the therapeutic ratio by safeguarding normal tissues without compromising tumor control [63]. Within

this strategy, amifostine exemplifies the protection approach, administered before radiotherapy to scavenge free radicals and prevent cellular injury. Palifermin represents the mitigation approach, given after irradiation but before overt injury, thereby reducing the severity of tissue damage. The treatment aspect primarily involves symptomatic and supportive interventions. An ideal radioprotector should selectively lower the risk of normal tissue complications while preserving antitumor efficacy. Clinically, amifostine has demonstrated significant benefits, such as reducing both acute and chronic xerostomia during head and neck cancer radiotherapy. However, its inherent toxicity and limited applicability in chemoradiotherapy remain concerns. Ultimately, pharmacological protection needs to be combined with advanced physical techniques, including intensity-modulated radiotherapy, to maximize treatment outcomes.

Highly conformal radiation therapy techniques, proton beam therapy, stereotactic radiosurgery, and IMRT can reduce the incidence of IRG. In clinical radiotherapy, techniques such as IMRT, proton therapy, and stereotactic radiosurgery have significantly reduced the impact of radiation on ocular structures by improving dose distribution accuracy. IMRT employs multiple beams of non-uniform intensity along with inverse planning systems to effectively spare critical organs at risk, such as the lens, retina, and optic nerve, thereby reducing the incidence of cataracts and retinopathy [64,65]. Proton therapy utilizes the Bragg peak to achieve controllable energy deposition, with a sharp dose drop-off beyond the target area, significantly minimizing radiation exposure to posterior ocular structures (e.g., the contralateral eyeball and optic nerve) and thus avoiding unnecessary irradiation to a certain extent [66]. Stereotactic radiosurgery (e.g., Gamma Knife, CyberKnife) delivers an ultra-high dose to the target in a single or a few sessions [67]. As the radiation beams converge from multiple angles, ocular tissues along the pathways receive only highly dispersed low-dose irradiation, thereby preserving functional structures to the greatest extent. Among these, IMRT can adjust the beam intensity according to the shape of the target, thereby reducing the dose to the trabecular meshwork and optic nerve [28,68].

Emerging therapeutic approaches

Recent advancements in ocular research offer promising directions for IRG treatment. Antiangiogenesis therapies, including ranibizumab and novel VEGF inhibitors, show potential in reducing neovascularization and managing NVG associated with radiation [69]. Beyond

conventional pharmacological interventions, gene therapy and stem cell therapy are emerging as transformative approaches [70,71]. Gene therapies targeting IOP regulation—by modulating pathways such as aquaporin 1, prostaglandin F_{2α} receptor, and RhoA/Rho kinase—hold promise for addressing both NVG and NTG [72]. Similarly, stem cell therapies using embryonic stem cells, induced pluripotent stem cells, and retinal progenitor cells offer the potential to regenerate damaged retinal ganglion cells and restore optic nerve function [73]. These innovative therapies, while still in experimental stages, underscore the importance of multidisciplinary collaboration in addressing IRG. As these approaches advance, they may provide effective, long-term solutions for patients, moving beyond symptom management to tackling the root causes of IRG.

Limitations and future directions

Despite the advancements highlighted in this review, significant limitations persist in the current evidence base. The heterogeneity of included studies—varying in radiation type, dose, and patient populations—limits the generalizability of findings. Additionally, the absence of long-term follow-up data hampers the ability to assess the durability of treatment effects. Standardized outcome measures are critically needed to facilitate cross-study comparisons and meta-analyses.

Future research should prioritize longitudinal studies with homogeneous populations and standardized methodologies to bridge these gaps. We give priority to precise shielding strategies as primary prevention, such as lens-sparing techniques during radiotherapy [74]. Investigations into the long-term efficacy of emerging therapies, such as gene and stem cell-based interventions, are also essential. Gene therapy must be administered prior to irreversible RGCs loss [75]. Furthermore, integrating advanced imaging and diagnostic tools, proposing biomarker-driven monitoring (e.g., aqueous VEGF levels combined with OCTA-detected microvascular changes), can enhance our understanding of early vascular changes in IRG, enabling proactive management strategies [76].

Conclusion

IRG poses a major clinical challenge owing to its complex pathophysiology and dose-dependent patterns. High-dose radiotherapy, particularly in uveal melanoma, predisposes to NVG, whereas chronic low-dose exposure is linked to NTG. Current options such as anti-VEGF therapy and surgical interventions provide partial benefit but lack standardization. Effective

management requires subtype-specific strategies, with anti-angiogenic approaches prioritized for NVG and neuroprotective measures for NTG. Preventive efforts should emphasize strict dose constraints to anterior ocular structures, protective shielding, and regular ocular monitoring. Future research must clarify molecular mechanisms and guide the development of targeted therapies. Addressing these gaps will improve both prevention and treatment, ultimately enhancing patient outcomes.

Author contribution statement

Qiwen Zhou: Data curation, Writing – original draft. Anqi Wu: Data curation, Writing – original draft. Xinyan Huai: Data curation, Writing – original draft. Chenghao Li and Heng Zhou: Conceptualization, Writing – review & editing.

Disclosure statement

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